

Marcin Copik

SERVERLESS · HPC · PHD RESEARCHER

ETH Zürich

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Summary

In my PhD research, I have been working on serverless programming models to bridge the gap between high-performance computing systems and cloud data centers. I developed tailored solutions for different levels of the FaaS computing stack: from computing and network devices to high-level optimizations, efficient system designs, and performance modeling.

Education

PhD in Computer Science

ETH ZÜRICH

April 2018 - March 2024

Zürich, Switzerland

- Thesis: High-Performance Serverless for HPC and Clouds
- Advisor: Prof. Torsten Hoefler

Master of Science (MSc) in Simulation Sciences

RWTH AACHEN

September 2014 - July 2017

Aachen, Germany

- Grade: 1.5. Interdisciplinary program. Major subject: High-Performance Computing
- Thesis: Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series
- Advisor: Prof. Paolo Bientinesi, Prof. Benjamin Berkels

Summer School in Mathematics

UNIVERSITY OF PERUGIA

August 2014

Perugia, Italy

- Courses: Stochastic Processes, Functional Analysis

Bachelor of Science (BSc) in Mathematics

SILESIA UNIVERSITY OF TECHNOLOGY

September 2012 - June 2014

Gliwice, Poland

- GPA: 4.6/5.0. Finished two of three years program

Bachelor of Science in Engineering (BSc) in Computer Science

SILESIA UNIVERSITY OF TECHNOLOGY

September 2010 - March 2014

Gliwice, Poland

- Grade 5(A). Major subject: Software Engineering
- Thesis: GPU-accelerated stochastic simulator engine for PRISM model checker
- Advisor: Prof. Tadeusz Czachorski

Experience

Postdoctoral Researcher

ETH ZÜRICH

Zürich, Switzerland

May 2024 -

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Assistant

ETH ZÜRICH

Zürich, Switzerland

April 2018 - April 2024

- Advising for Bachelor and Master thesis projects.
- Conducting interviews for PhD and PostDoc candidates.
- Teaching assistant for Bachelor and Master courses.

Research Intern

MICROSOFT

Redmond, WA, USA

June - October 2019

- Analyzing microarchitectural implications of serverless workloads.
- Supervisor: Bobbie Manne.

Mentor

GOOGLE SUMMER OF CODE

Organization: The STE||AR Group.

2017, 2018

- Mentoring students working on HPX.
- Students: Ajai V George, Gabriel Laberge (co-mentored).

Student Research Assistant

RWTH AACHEN, HIGH-PERFORMANCE AND AUTOMATIC COMPUTING

- Benchmarking linear algebra frameworks.
- Supervisor: Prof. Paolo Bientinesi.

Aachen, Germany

2016 - December 2017

Research Assistant

LOUISIANA STATE UNIVERSITY, STE||AR GROUP

- Integrating single-source GPU programming in HPX.
- Supervisor: Prof. Hartmut Kaiser.

Baton Rouge, LA, USA

April 2016 - August 2016

Student Research Assistant

JÜLICH SUPERCOMPUTING CENTRE

- Developing tools for performance analysis of parallel applications at Scalasca.
- Supervisor: Dr Pavel Saviankou.

Jülich, Germany

October 2014 - March 2016

Software Engineer

GOOGLE SUMMER OF CODE

- Integrating single-source GPU programming in HPX.
- Supervisor: Dr Hartmut Kaiser.

Organization: The STE||AR Group

2015

Software Engineer

GOOGLE SUMMER OF CODE

- Improving statistical model checking.
- Supervisor: Dr Vojtěch Forejt, Dr Dave Parker.

Organization: PRISM model checker

2014

Student Research Assistant

THE INSTITUTE OF THEORETICAL AND APPLIED INFORMATICS

- Implementing GPU simulator of Markov Chains.
- Supervisors: Dr Mateusz Nowak, Dr Artur Rataj.

Gliwice, Poland

2012 - 2013

Student Research Assistant

SILESIA UNIVERSITY OF TECHNOLOGY

- Implementing algorithms for registration of respiratory motion.
- Supervisor: Dr Dominik Spinczyk.

Gliwice, Poland

2012 - 2013

Honors & Awards

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| 2024 | Best Research Poster Award , ACM/IEEE Supercomputing 2024 |
| 2024 | SIGHPC Travel Grant , awarded for travel to ACM/IEEE Supercomputing 2024. |
| 2023 | SIGHPC Travel Grant , awarded for travel to ACM/IEEE Supercomputing 2023. |
| 2022 | ACM/IEEE George Michael Memorial HPC Fellowship , awarded for contributions into high-performance serverless. |
| 2022 | Gold Medal at the ACM Student Research Competition , ACM/IEEE Supercomputing 2022 |
| 2022 | AWS Cloud Credit for Research Application |
| 2022 | Google Cloud Research Credits |
| 2021 | Microsoft Research PhD Fellowship , awarded for the 2021/2022 academic year. |
| 2019 | Gold Medal at the ACM Student Research Competition , ACM/IEEE Supercomputing 2019 |

Peer-reviewed Publications

A Priori Loop Nest Normalization: Automatic Loop Scheduling in Complex Applications

TRÜMPER L., SCHAAD P., ATEs B., CALOTOIU A., **COPIK M.**, HOEFLE T.

CGO

2025

Protocol Buffer Deserialization DPU Offloading in the RPC Datapath

FRANTZ R., GARCIA J., **COPIK M.**, MONROY I., OLMOS J., BLOCH G., DI GIROLAMO S.

IXPUG Workshop @ Supercomputing

2024

Process-as-a-Service: Unifying Elastic and Stateful Clouds with Serverless Processes

COPIK M., CALOTOIU A., GYORGY R., BÖHRINGER R., BRUNO R., HOEFLE T.

ACM SoCC

2024

- Acceptance Rate 30.1% (63/209)

FaaSKeeper: Learning from Building Serverless Services with ZooKeeper as an Example

COPIK M., CALOTOIU A., ZHOU P., TARANOV K., HOEFLE T.

ACM HPDC

2024

- Acceptance Rate 17.3% (26/150)

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing

HOEFLE T., **COPIK M.**, BECKMAN P., JONES A., FOSTER I., PARASHAR M., REED D., TROYER M., SCHULTHESS T., ERNST D.,
DONGARRA J.

IEEE CISE

2024

Software Resource Disaggregation for HPC with Serverless Computing

COPIK M., CHRAPEK M., SCHMID L., CALOTOIU A., HOEFLE T.

- Acceptance Rate 26.1% (88/337)

IEEE IPDPS

2024

User-guided Page Merging for Memory Deduplication in Serverless Systems

QIU W., **COPIK M.**, WANG Y., CALOTOIU A., HOEFLE T.

- Acceptance Rate 17.5% (92/526)

IEEE Big Data

2023

FMI: Fast and Cheap Message Passing for Serverless Functions

COPIK M., BÖHRINGER R., CALOTOIU A., HOEFLE T.

- Acceptance Rate 29.4% (40/136)

ACM ICS

2023

rFaaS: Enabling High Performance Serverless with RDMA and Leases

COPIK M., TARANOV K., CALOTOIU A., HOEFLE T.

- Acceptance Rate 25.7% (95/369)

IPDPS

2023

Performance-Detective: Automatic Deduction of Cheap and Accurate Performance Models

SCHMID L., **COPIK M.**, CALOTOIU A., WERLE D., REITER A., SELZER M., KOZIOLEK A., HOEFLE T.

- Acceptance Rate 24.2% (39/161)

ACM ICS

2022

MOM: Matrix Operations in MLIR

CHELINI L., BARTHELS H., BIENTINESI P., **COPIK M.**, GROSSER T., SPAMINATO D.

IMPACT

2022

Work-stealing Prefix Scan: Addressing Load Imbalance in Large-scale Image Registration

COPIK M., GROSSER T., HOEFLE T., BIENTINESI P., BERKELS B.

IEEE TPDS

2021

SeBS: A Serverless Benchmark Suite for Function-as-a-Service Computing

COPIK M., KWASNIEWSKI G., BESTA M., PODSTAWSKI M., HOEFLE T.

- Acceptance Rate 31% (33/107)

ACM/IFIP Middleware

2021

Extracting Clean Performance Models from Tainted Programs

COPIK M., CALOTOIU A., GROSSER T., WICKI N., WOLF F., HOEFLE T.

- Acceptance Rate 21% (31/150)

ACM PPOPP

2021

GraphMineSuite: Enabling High-Performance and Programmable Graph Mining Algorithms with Set Algebra

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**]

VLDB

2021

SISA: Set-Centric Instruction Set Architecture for Graph Mining on Processing-in-Memory Systems

BESTA M. [AND 18 OTHERS, INCLUDING **COPIK M.**]

IEEE MICRO

2021

The Generalized Matrix Chain Algorithm

BARTHELS H., **COPIK M.**, BIENTINESI P.

- Acceptance Rate 28.6% (30/105)

CGO

2018

Using SYCL as an Implementation Framework for HPX.Compute

COPIK M., KAISER H.

DHPCC++ Workshop, IWOCL

2017

A GPGPU-based Simulator for Prism: Statistical Verification of Results of PMC

COPIK M., RATAJ A., WOŻNA-SZCZĘŚNIAK B.

CS&P

2016

Methods for abdominal respiratory motion tracking

SPINCYK D., KARWAN A., **COPIK M.**

Computer Aided Surgery

2014

Presentations and Talks

XaaS: Acceleration as a Service to Enable Productive High-Performance Cloud Computing

PANEL MODERATOR AT CANOPIE-HPC WORKSHOP @ ACM/IEEE SUPERCOMPUTING 2024

November 2024

Serverless HPC: Challenges, Opportunities, and Future Prospects for Accelerated Cloud Computing

PANEL MODERATOR AT ACM/IEEE SUPERCOMPUTING 2024

November 2024

MIGNificent: Fast, Isolated, and GPU-Enabled Serverless Functions ACM/IEEE SUPERCOMPUTING 2024 POSTER	November 2024
Benchmarking Serverless with SeBS: Past, Present, and Future THIRD INTERNATIONAL WORKSHOP ON SERVERLESS COMPUTING EXPERIENCE 2024	June 2024
High Performance Serverless for HPC and Cloud INVITED TALK, INTELLIGENT SERVERLESS AND CLOUD APPLICATIONS SYMPOSIUM, ZURICH UNIVERSITY OF APPLIED SCIENCES.	June 2024
Evaluating FaaS Systems with the Serverless Benchmark Suite SeBS SEATED WORKSHOP ON SERVERLESS AT THE EDGE, HPDC 2024	June 2024
Cppless: Productive and Performant Serverless Programming in C++ LIGHTNING TALK, LLVM-HPC AT ACM/IEEE SUPERCOMPUTING 2023.	November 2023
High Performance Serverless for HPC and Clouds POSTER PRESENTATION AT DOCTORAL SHOWCASE, SC 2023.	November 2023
Serverless As a Bridge Between HPC and Clouds INVITED TALK, AWS CLOUD FOR RESEARCH AT ETH.	May 2023
Serverless As a Bridge Between HPC and Clouds INVITED TALK, 5TH WORKSHOP ON PARALLEL AI AND SYSTEMS FOR THE EDGE (PAISE), IPDPS 2023.	May 2023
Serverless As a Bridge Between HPC and Clouds POSTER PRESENTATION AT PHD FORUM, IPDPS 2023.	May 2023
Software Resource Disaggregation for HPC with Serverless Computing ACM/IEEE SUPERCOMPUTING 2022 POSTER, GOLD MEDAL AT THE ACM STUDENT RESEARCH COMPETITION.	November 2022
Software Resource Disaggregation for HPC with Serverless Computing SUPERCOMPCLOUD AT ACM/IEEE SUPERCOMPUTING 2022.	November 2022
Interactive Computing with Serverless Functions in rFaaS LIGHTNING TALK, URGENTHPC AT ACM/IEEE SUPERCOMPUTING 2022.	November 2022
Extracting Clean Performance Models from Tainted Programs” SIAM CONFERENCE ON PARALLEL PROCESSING FOR SCIENTIFIC COMPUTING (PP22) MINISYMPOSIUM	February 2022
perf-taint: Taint Analysis for Automatic Many-Parameter Performance Modeling ACM/IEEE SUPERCOMPUTING 2019 POSTER, GOLD MEDAL AT THE ACM STUDENT RESEARCH COMPETITION.	November 2019
Parallel Prefix Algorithms for the Registration of Arbitrarily Long Electron Micrograph Series ACM/IEEE SUPERCOMPUTING 2017 POSTER, ACM STUDENT RESEARCH COMPETITION.	November 2017
HPX and GPU-parallelized STL C++NOW 2016 CONFERENCE.	May 2016


Skills

Programming	Experienced: C++, Python, Java Familiar: Matlab, Julia, Mathematica, R, Pascal, x86 ASM
Technologies	MPI, OpenMP, LLVM, OpenCL, SYCL, C++AMP, Docker, Kubernetes
Tools	Git, SVN, Mercurial, CMake, autotools, SLURM
Experience	serverless computing, parallel programming, cloud computing, performance modeling, GPU programming, model checking
Languages	English, German, Polish

Service

2025	Workshop on Parallel AI and Systems for the Edge (PAISE)	Organizing committee
2025	EuroSys	Shadow PC
2025	Accelerated HPC in the Cloud-Edge Continuum (AHPC3) @ PDP	Program Committee
2025	IEEE IPDPS	Program Committee
2024	Journal of Systems Architecture	Reviewer
2024	Workshop on Parallel AI and Systems for the Edge (PAISE)	Organizing committee
2023	ACM/IEEE Supercomputing	Student Volunteer
2022	ACM/IEEE Supercomputing	Student Volunteer
2022	International Journal of High Performance Computing Applications	Reviewer
2020	LLVM-HPC Workshop	Reviewer
2019	ISC High Performance	Reviewer

Teaching

Fall 2024	Information Systems for Engineers	ETH Zürich
Spring 2024	Parallel Programming	ETH Zürich
Fall 2023	Big Data	ETH Zürich
Spring 2023	Parallel Programming	ETH Zürich
Spring 2022	Parallel Programming	ETH Zürich
Fall 2021	Information Systems for Engineers	ETH Zürich
Spring 2021	Parallel Programming	ETH Zürich
Fall 2020	Compiler Design	ETH Zürich
Spring 2020	Parallel Programming	ETH Zürich
Fall 2019	Design of Parallel and High-Performance Computing	ETH Zürich
Spring 2019	Parallel Programming	ETH Zürich
Fall 2018	Numerical Methods for Computational Science and Engineering	ETH Zürich

Students

Andrea Jiang	Co-supervised Master Thesis: Serverless and Cloud Runtimes for Graph-of-Thoughts	2024, ETH Zürich
Entiol Liko	Semester Project: Serverless Co-location with ML	2024, ETH Zürich
Oana Rosca	Semester Project: Long-Term Serverless Performance Variability	2024, ETH Zürich
Constantin Dragancea	Master Thesis: Adoption and evolution of C++ in HPC Applications	2024, ETH Zürich
Prajin Khadka	Co-supervised Google Summer of Code Student: Expanded serverless benchmarks	2024, GSoC
Syed Mujtaba	Google Summer of Code Student: Using serverless ZooKeeper in Apache projects	2024, GSoC
Abhishek Kumar	Co-supervised Google Summer of Code Student: New serverless benchmarks	2024, GSoC
Matt Nappo	Co-supervised Google Summer of Code Student: Libfabric Implementation of rFaaS	2023, GSoC
Boyan Zhou	Master Thesis: Adoption and evolution of C++ in HPC Applications	2023, ETH Zürich
Gyorgy Rethy	Master Thesis: Process-as-a-Service computing on modern serverless platforms	2022, ETH Zürich
Laurin Brandner	Master Thesis: Serverless workflows benchmarking	2022, ETH Zürich
Lukas Möller	Bachelor Thesis: Serverless C++ Executor	2022, ETH Zürich
Malte Wächter	Bachelor Thesis: Profiling and optimizations of serverless functions	2022, ETH Zürich
Qiu Wei	Master Thesis: Serverless memory deduplication	2022, ETH Zürich
Lukas Tobler	Master Thesis: Serverless GPU functions	2022, ETH Zürich
Arnet Colin	Bachelor Thesis: Verification of representativeness of benchmarking suite	2021, ETH Zürich
Roman Böhringer	Master Thesis: Serverless collectives.	2021, ETH Zürich
Emir İşman	Bachelor Thesis: FaaSStest collectives: reliable communication in serverless world	2021, ETH Zürich
Konrad Handrick	Co-supervised Bachelor Thesis: Offloading serverless with sPIN	2021, ETH Zürich
Tobias Lüscher	Bachelor Thesis: TaintImpact: Taint-Based Change Impact Analysis	2021, ETH Zürich
Siegfried Hartogs	Bachelor Thesis: Code-driven Language Development: Framework for Analysis of C/C++ Open-Source Projects	2021, ETH Zürich
Lukas Gygi	Bachelor Thesis: CppBuild: Large-Scale, Automatic Build System for Open Source C++ Repositories	2021, ETH Zürich
Nicolas Wicki	Bachelor Thesis: Control Flow Taint Analysis for Performance Modeling in LLVM	2020, ETH Zürich

Philipp Bomatter

Co-supervised Bachelor Thesis: Towards Extreme-Scale Cache Coherence Protocols and Simulations

2019, *ETH Zürich*

Gabriel Laberge

Co-supervised Google Summer of Code Student: Alternative smart executors

2018, *GSoC*

Ajai V George

Google Summer of Code Student: Work on Parallel Algorithms

2017, *GSoC*